LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Practice Pack: Motion

Chapter / Topic: Topic-based: use with whichever Class 9 book your students follow

For teacher use only. Do not distribute to students.

LEVEL 1: CONCEPTS

Q1. Define displacement. Can the displacement of a body be zero when the distance it has covered is not zero? Give an example.

Answer: Displacement is the shortest distance from the initial to the final position, in a stated direction. Yes: if a body returns to its starting point, for example one full round of a circular track, its displacement is zero but the distance covered is not.

Q2. A body covers equal distances in equal intervals of time. What kind of motion is this?

Answer: Uniform motion.

Q3. What does the slope of a distance–time graph tell us?

Answer: The speed of the body. A steeper slope means a higher speed.

Q4. What does the area under a velocity–time graph give?

Answer: The displacement (equal to the distance for motion in one direction along a straight line).

Q5. Is uniform circular motion accelerated? Explain.

Answer: Yes. The speed is constant, but the direction of motion changes at every point, so the velocity changes and the motion is accelerated.

LEVEL 2: NUMERICALS

Q6. A car travels 150 km in 3 hours. Find its average speed in km/h and in m/s.

Answer: Average speed = 150 km/3 h = 50 km/h = 50/3.6 ≈ 13.9 m/s.

Q7. An athlete completes one round of a circular track of diameter 200 m in 40 s. What distance has he covered, and what is his displacement, at the end…

Answer: Time = 140 s = 3.5 rounds. Circumference = (22/7) × 200 ≈ 628.6 m, so distance = 3.5 × 628.6 = 2200 m. After 3.5 rounds he is at the opposite end of a diameter, so displacement = 200 m.

Q8. A bus slows down from 80 km/h to 60 km/h in 5 s. Find its acceleration.

Answer: u = 80/3.6 ≈ 22.2 m/s, v = 60/3.6 ≈ 16.7 m/s. a = (v – u)/t = (16.7 – 22.2)/5 ≈ –1.1 m/s^2. The negative sign shows retardation.

Q9. A train starting from rest reaches a speed of 72 km/h in 5 minutes. Find its acceleration and the distance it travels in this time.

Answer: u = 0, v = 72 km/h = 20 m/s, t = 300 s. a = 20/300 ≈ 0.067 m/s^2. s = ½at^2 = ½ × (1/15) × 300^2 = 3000 m = 3 km.

Q10. A car accelerates uniformly from 18 km/h to 36 km/h in 5 s. Find (i) the acceleration and (ii) the distance covered in that time.

Answer: u = 5 m/s, v = 10 m/s. (i) a = (10 – 5)/5 = 1 m/s^2. (ii) s = ut + ½at^2 = 5 × 5 + ½ × 1 × 25 = 37.5 m.

Q11. A car moving at 20 m/s is brought to rest with a uniform retardation of 4 m/s^2. Find the time taken and the stopping distance.

Answer: t = (v – u)/a = (0 – 20)/(–4) = 5 s. s = (v^2 – u^2)/2a = (0 – 400)/(–8) = 50 m.

Q12. A satellite moves in a circular orbit of radius 42,250 km and takes 24 hours to go around the Earth once. Find its speed.

Answer: $v = 2\pi r/t = (2 \times 3.14 \times 42{,}250\ \text{km})/24\ \text{h} \approx 11{,}060\ \text{km/h} \approx 3.07\ \text{km/s}$.

LEVEL 3: CHALLENGE (FOUNDATION)

Q13. A car covers the first half of a journey at 40 km/h and the second half at 60 km/h. Find its average speed for the whole journey. (Hint: it is NOT 50…

Answer: Let each half be d km. Time = $d/40 + d/60 = 5d/120 = d/24$ h. Average speed = $2d \div (d/24) = 48$ km/h. (For equal distances, average speed = $2v_1v_2/(v_1 + v_2)$.)

Q14. Car A starts from rest and accelerates at 2 m/s^2 for 5 s. Car B starts from rest and accelerates at 1 m/s^2 for 10 s. Which car ends with the greater…

Answer: A: $v = 2 \times 5 = 10$ m/s, $s = ½ \times 2 \times 25 = 25$ m. B: $v = 1 \times 10 = 10$ m/s, $s = ½ \times 1 \times 100 = 50$ m. Both end at the same speed (10 m/s), but B covers twice the distance.

Q15. A velocity–time graph is a straight line rising from 0 to 20 m/s in the first 4 s, then horizontal at 20 m/s for the next 6 s. Find the acceleration…

Answer: $a = 20/4 = 5\ m/s^2$. Distance = area under the graph = triangle ($½ \times 4 \times 20 = 40$ m) + rectangle ($6 \times 20 = 120$ m) = 160 m.